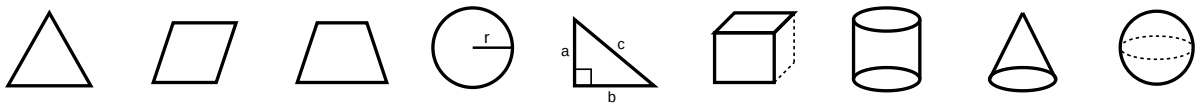
Grade 8 Mathematics Reference Sheet

On the real STAAR test, these formulas are provided for you. Learn where each one is so you can find it fast. (This replicates the key formulas from the STAAR Grade 8 Mathematics Reference Materials.)

Shapes at a glance



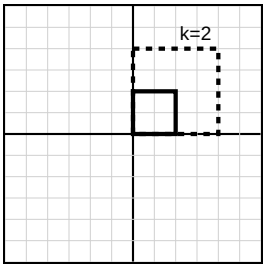
Triangle · Parallelogram · Trapezoid · Circle · Right triangle · Prism · Cylinder · Cone · Sphere

Perimeter & Circumference	Area
Circle: $\mathbf{C = 2\pi r}$ or $\mathbf{C = \pi d}$	Triangle: $\mathbf{A = \frac{1}{2}bh}$ Rectangle / Parallelogram: $\mathbf{A = bh}$ Trapezoid: $\mathbf{A = \frac{1}{2}(b_1 + b_2)h}$ Circle: $\mathbf{A = \pi r^2}$
Surface Area (B = base area, P = base perimeter)	**Volume** (B = base area)
Prism — Lateral: $\mathbf{S = Ph}$ · Total: $\mathbf{S = Ph + 2B}$ Cylinder — Lateral: $\mathbf{S = 2\pi rh}$ · Total: $\mathbf{S = 2\pi rh + 2\pi r^2}$	Prism / Cylinder: $\mathbf{V = Bh}$ Cone: $\mathbf{V = \frac{1}{3}Bh}$ Sphere: $\mathbf{V = \frac{4}{3}\pi r^3}$
Algebra & Graphing	**Personal Financial Literacy**
Slope: $\mathbf{m = (y_2 - y_1) / (x_2 - x_1)}$ Slope-intercept form: $\mathbf{y = mx + b}$ Direct variation: $\mathbf{y = kx}$ Pythagorean theorem: $\mathbf{a^2 + b^2 = c^2}$	Simple interest: $\mathbf{I = Prt}$ Compound interest: $\mathbf{A = P(1 + r/n)^{nt}}$
Useful Constants & Conversions	
$\mathbf{\pi \approx 3.14}$ (or $22/7$) · Metric: 1 m = 100 cm = 1000 mm; 1 km = 1000 m · Customary: 1 ft = 12 in; 1 yd = 3 ft; 1 mi = 5,280 ft	

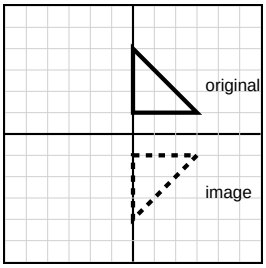
Strategy: You do **not** need to memorize these — but you must know they exist and how to use them. Every worked example that uses a formula shows you which one and why.

Strand 3 · Geometry and Measurement

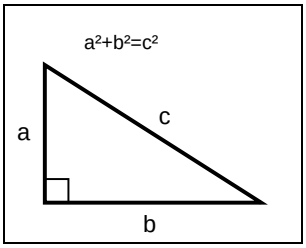
Key Figures for This Strand



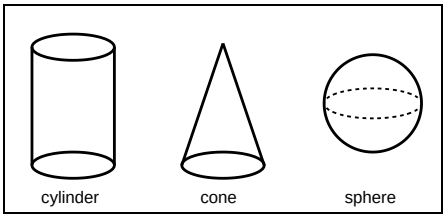
Dilation (8.3C, 8.10D)



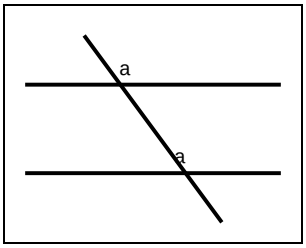
Reflection (8.10C)



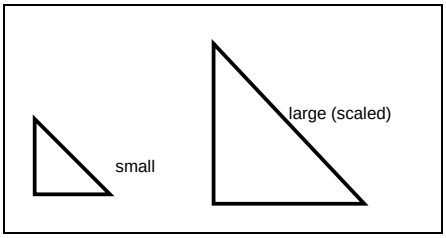
Pythagorean (8.7C)



Volume solids (8.7A)



Transversal angles (8.8D)



Similar figures (8.3A)

This strand (about 13 of 40 questions) covers **dilations, transformations, angles, volume, surface area, and the Pythagorean theorem.** Keep the **reference sheet** open — most of these use a formula.

Standards covered:

- ○ **8.3A** — Ratio of corresponding sides of similar shapes is proportional. ○ **8.3B** — Compare a shape and its dilation on a plane. ★ **8.3C** — Algebraic rule for a scale factor (origin center).
- ○ **8.6A** — Volume formula V = Bh of a cylinder. ○ **8.6C** — Explain the Pythagorean theorem with models.
- ★ **8.7A** — Volume of cylinders, cones, spheres. ★ **8.7B** — Surface area of prisms and cylinders. ★ **8.7C** — Pythagorean theorem and its converse. ○ **8.7D** — Distance between two points.
- ○ **8.8D** — Angle sums, exterior angles, transversal angles, AA similarity.
- ○ **8.10A** — Orientation/congruence of transformations. ○ **8.10B** — Which preserve congruence. ★ **8.10C** — Rules for translations, reflections, rotations (90/180/270/360). ○ **8.10D** — Effect of a dilation on length and area.

Part A · Dilations and Similarity (8.3A ○, 8.3B ○, 8.3C ★, 8.10D ○)

A **dilation** resizes a figure by a **scale factor k** from a center. With the **origin** as center: **(x, y) → (kx, ky).**

- **8.3A / 8.3B — Similar shapes.** A figure and its dilation are **similar**: corresponding sides are proportional (×k) and angles stay the **same.**